

Structural Syzygies and an Infinite-Dimensional Family of Counterexamples to the Three-Dimensional Jacobian Conjecture

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Abstract

The Jacobian Conjecture, proposed by Keller in 1939, asserts that any polynomial mapping $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ with a non-vanishing constant Jacobian determinant is a global polynomial automorphism. Listed by Smale as one of the mathematical problems for the 21st century, the conjecture has stood for nearly nine decades. Recently, Levent Alpöge (with contributions from Akhil Mathew) announced an explicit polynomial mapping $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ with a constant Jacobian determinant of -2 that is not injective, thereby disproving the conjecture for all dimensions $n \geq 3$. In this paper, we generalize Alpöge's counterexample to an infinite-dimensional affine family of maps $F_{t(u)}$ parameterized by an arbitrary polynomial $t(u) \in \mathbb{C}[u]$. We establish two polynomial syzygies in the coordinate ring satisfied by the components of this generalized mapping, which are governed by a Bézout identity on the auxiliary variable $u = 1 + xy$. We show that these relations reduce the fiber analysis to a quadratic equation in one variable, completely bypassing Gröbner basis computations. We also prove that the Jacobian determinant is constant and identically equal to -2 for the entire family, and that the (x, y) -locus of the fiber over $(-\frac{1}{4}, 0, 0)$ is completely rigid, containing exactly three points for any choice of $t(u)$. Finally, we show that the parameterization yields infinitely many inequivalent counterexamples under the action of affine automorphisms.

1 Introduction

Let $\mathbb{C}[x_1, \dots, x_n]$ denote the polynomial ring in n variables over the complex numbers. A polynomial mapping $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is given by $F = (F_1, \dots, F_n)$ where each $F_i \in \mathbb{C}[x_1, \dots, x_n]$. The Jacobian matrix of F is defined as:

$$J_F(x) = \left(\frac{\partial F_i}{\partial x_j} \right)_{1 \leq i, j \leq n}.$$

The Jacobian Conjecture can be formally stated as follows:

Definition 1.1 (Jacobian Conjecture). Let $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be a polynomial mapping. If $\det J_F(x) = c$ for some constant $c \in \mathbb{C} \setminus \{0\}$, then F is a global bijection, and its inverse mapping $F^{-1} : \mathbb{C}^n \rightarrow \mathbb{C}^n$ is also a polynomial mapping.

Keller originally posed this problem in 1939 for integer polynomial mappings [3]. In 1982, Bass, Connell, and Wright [2], and independently Yagzhev [5], showed that the conjecture can be reduced to the case of cubic homogeneous mappings of the form $F(x) = x - H(x)$, where each component of H is homogeneous of degree 3.

While the real analogue of the conjecture (for mappings $\mathbb{R}^2 \rightarrow \mathbb{R}^2$) was disproved by Sergey Pinchuk in 1994 via a degree-25 polynomial map [4], the complex version remained open in all dimensions $n \geq 2$.

Recently, Levent Alpöge [1], using computational search strategies assisted by the Anthropic Claude Fable 5 model, announced a counterexample in dimension $n = 3$. By standard stabilization arguments, this settled the conjecture in the negative for all $n \geq 3$. The two-variable case ($n = 2$) remains a major open problem.

1.1 Main Contributions

Our primary contributions in this paper are:

- We generalize Alpöge’s counterexample mapping to an infinite-dimensional affine family of maps $F_{t(u)}$ parameterized by an arbitrary polynomial $t(u) \in \mathbb{C}[u]$.
- We establish that these mappings are governed by a polynomial Bézout identity on the auxiliary variable $u = 1 + xy$, exposing two algebraic syzygies.
- We prove that the fiber characterization remains completely rigid in the (x, y) -locus under this generalization, resolving to a 3-point collision for any choice of $t(u)$.
- We present a direct symbolic proof of the Jacobian determinant’s constancy, showing that it remains identically -2 for all members of the family.
- We prove that varying the degree of $t(u)$ generates infinitely many inequivalent counterexamples under the action of affine automorphisms of $\mathbb{C}^3$.

1.2 Related Work and Context

Following Alpöge’s announcement, several independent verifications emerged, including formalizations in the Lean interactive theorem prover [7] and algebraic investigations of the mapping’s properties [6]. Zihan Zhang’s preprint [6] focuses on the disproof of the related Image and Vanishing conjectures in dimension three. While computer algebra systems (CAS) verify the Jacobian determinant, they do not immediately illuminate the underlying algebraic structure. Lean verifies correctness mathematically but behaves as a proof checking oracle. The present paper complements these approaches by providing a structural explanation of *why* the counterexample works, and presenting a broad classification under a parameterized family.

2 The Infinite-Dimensional Mapping Family

To analyze the mapping family, we introduce the auxiliary variable:

$$u(x, y) = 1 + xy.$$

For any polynomial $t(u) \in \mathbb{C}[u]$, we define the parameterizing polynomials:

$$\varphi(u) = 1 + 3u + u^2 t(u), \tag{1}$$

$$\psi(u) = 5 - 3u - (u - 1)^2 t(u). \tag{2}$$

These polynomials satisfy the fundamental Bézout identity:

$$(u - 1)^2 \varphi(u) + u^2 \psi(u) = u + 1. \tag{3}$$

Indeed, expanding the left-hand side yields:

$$\begin{aligned} (u - 1)^2 (1 + 3u + u^2 t(u)) + u^2 (5 - 3u - (u - 1)^2 t(u)) &= (u - 1)^2 (1 + 3u) + u^2 (5 - 3u) \\ &= (u^2 - 2u + 1)(3u + 1) + 5u^2 - 3u^3 \\ &= (3u^3 - 5u^2 + u + 1) + 5u^2 - 3u^3 \\ &= u + 1. \end{aligned}$$

Using these functions, we define the polynomial mapping $F_{t(u)} = (P, Q, R) : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ by:

$$P(x, y, z) = u^3 z + y^2 u \varphi(u), \tag{4}$$

$$Q(x, y, z) = y + 3xu^2 z + 3xy^2 \varphi(u), \tag{5}$$

$$R(x, y, z) = x (\psi(u) - x^2 z). \tag{6}$$

Alpöge’s original counterexample corresponds to the special case $t(u) \equiv 0$.

2.1 Discovery and Algebraic Syzygies

Since the coordinate functions P and Q depend linearly on z , one can eliminate z directly from the expressions. In the coordinate ring $\mathbb{C}[x, y, z]$, this elimination yields a relationship between P, Q , and the variables x, y . Specifically, we find two algebraic syzygies that relate the coordinates of the map:

Proposition 2.1. *Let $F_{t(u)} = (P, Q, R)$ be the polynomial mapping defined above. Then the following algebraic identities hold in $\mathbb{C}[x, y, z]$:*

$$u(Q - y) = 3xP, \quad (7)$$

$$x^3P + u^3R = xu(u + 1). \quad (8)$$

Proof. To prove (7), we substitute (4) into the right-hand side:

$$3xP = 3x(u^3z + y^2u\varphi(u)) = 3xu^3z + 3xy^2u\varphi(u).$$

Now we compute the left-hand side using (5):

$$u(Q - y) = u(3xu^2z + 3xy^2\varphi(u)) = 3xu^3z + 3xy^2u\varphi(u).$$

Thus, $u(Q - y) = 3xP$ holds identically.

To prove (8), we substitute (4) and (6) into the left-hand side:

$$\begin{aligned} x^3P + u^3R &= x^3(u^3z + y^2u\varphi(u)) + u^3(x(\psi(u) - x^2z)) \\ &= x^3u^3z + x^3y^2u\varphi(u) + xu^3\psi(u) - x^3u^3z \\ &= x(xy)^2u\varphi(u) + xu^3\psi(u). \end{aligned}$$

Since $xy = u - 1$, we have:

$$\begin{aligned} x^3P + u^3R &= x(u - 1)^2u\varphi(u) + xu^3\psi(u) \\ &= xu[(u - 1)^2\varphi(u) + u^2\psi(u)]. \end{aligned}$$

Applying the Bézout identity (3):

$$x^3P + u^3R = xu(u + 1).$$

This completes the proof. $\square$

2.2 Relation to Elimination Ideals and Syzygy Modules

Let $R' = \mathbb{C}[x, y, z]$. Consider the components $\{P, Q - y, R\}$ as elements of the ring R' . The identity (7) implies that the vector $(3x, -u, 0)^T \in (R')^3$ lies in the syzygy module of $\{P, Q - y, R\}$. Similarly, identity (8) modulo the ideal $\langle xu(u + 1) \rangle$ shows that $(x^3, 0, u^3)^T$ is a syzygy modulo the subring relations.

More formally, let $I = \langle P - P(x, y, z), Q - Q(x, y, z), R - R(x, y, z) \rangle$ be the ideal in $\mathbb{C}[x, y, z, P, Q, R]$. The syzygies (7) and (8) represent explicit elements of the elimination ideal $I \cap \mathbb{C}[x, y, P, Q, R]$, showing that z can be eliminated through direct algebraic substitution.

3 Direct Symbolic Proof of the Jacobian Determinant

In this section, we show that the Jacobian determinant of $F_{t(u)}$ is a constant. We note the differentiation rules for $u(x, y) = 1 + xy$:

$$\frac{\partial u}{\partial x} = y, \quad \frac{\partial u}{\partial y} = x.$$

Theorem 3.1. *For any polynomial $t(u) \in \mathbb{C}[u]$, the Jacobian determinant of the mapping $F_{t(u)} = (P, Q, R)$ is identically equal to -2 on $\mathbb{C}^3$.*

Proof. Let J_F be the Jacobian matrix of F :

$$J_F = \begin{pmatrix} P_x & P_y & P_z \\ Q_x & Q_y & Q_z \\ R_x & R_y & R_z \end{pmatrix}.$$

We compute the partial derivatives of the coordinate functions. The z -derivatives are:

$$P_z = u^3, \quad Q_z = 3xu^2, \quad R_z = -x^3.$$

The remaining partial derivatives are linear in z . We write them in the form $M_v = M_{v,1}z + M_{v,0}$ for $M \in \{P, Q, R\}$ and $v \in \{x, y\}$:

$$\begin{aligned} P_x &= (3yu^2)z + y^2u_x\varphi(u) + y^2u\varphi'(u)u_x = (3yu^2)z + y^3(\varphi(u) + u\varphi'(u)), \\ P_y &= (3xu^2)z + 2yu\varphi(u) + y^2u_y\varphi(u) + y^2u\varphi'(u)u_y = (3xu^2)z + y((2u + xy)\varphi(u) + uxy\varphi'(u)), \\ Q_x &= (3u(3u - 2))z + 3y^2\varphi(u) + 3xy^2\varphi'(u)u_x = (3u(3u - 2))z + 3y^2(\varphi(u) + xy\varphi'(u)), \\ Q_y &= (6x^2u)z + 1 + 6xy\varphi(u) + 3xy^2\varphi'(u)u_y = (6x^2u)z + 1 + 6xy\varphi(u) + 3x^2y^2\varphi'(u), \\ R_x &= (-3x^2)z + \psi(u) + x\psi'(u)u_x = (-3x^2)z + \psi(u) + xy\psi'(u), \\ R_y &= x^2\psi'(u). \end{aligned}$$

We evaluate $\det J_F$ via cofactor expansion along the third column:

$$\det J_F = u^3(Q_x R_y - Q_y R_x) - 3xu^2(P_x R_y - P_y R_x) - x^3(P_x Q_y - P_y Q_x).$$

Since P_z, Q_z, R_z are independent of z , and all other partial derivatives are at most linear in z , the determinant $\det J_F$ is a polynomial in z of degree at most 2. We write:

$$\det J_F = Az^2 + Bz + C.$$

We now prove that the coefficients A and B vanish identically.

3.1 Vanishing of the z^2 Coefficient

The term A receives contributions from the z^2 -components of each of the three terms in $\det J_F$.

1. From $u^3(Q_x R_y - Q_y R_x)$: Since R_y is independent of z , the term $Q_x R_y$ has no z^2 component. The term $Q_y R_x$ has z^2 coefficient $Q_{y,1}R_{x,1} = (6x^2u)(-3x^2) = -18x^4u$. Thus, the first term contributes $-u^3(-18x^4u) = 18x^4u^4$.
2. From $-3xu^2(P_x R_y - P_y R_x)$: Similarly, $P_x R_y$ has no z^2 component. The term $P_y R_x$ has z^2 coefficient $P_{y,1}R_{x,1} = (3xu^2)(-3x^2) = -9x^3u^2$. The contribution is $-3xu^2(-(-9x^3u^2)) = -27x^4u^4$.
3. From $-x^3(P_x Q_y - P_y Q_x)$: Here, both terms contain z^2 components.

$$P_{x,1}Q_{y,1} - P_{y,1}Q_{x,1} = (3yu^2)(6x^2u) - (3xu^2)(3u(3u - 2)) = 18x^2yu^3 - 9xu^3(3u - 2).$$

Substituting $x^2y = x(xy) = x(u - 1)$:

$$18x(u - 1)u^3 - 9xu^3(3u - 2) = 9xu^3[2(u - 1) - (3u - 2)] = -9xu^4.$$

Thus, the contribution is $-x^3(-9xu^4) = 9x^4u^4$.

Summing these contributions, we obtain:

$$A = 18x^4u^4 - 27x^4u^4 + 9x^4u^4 = 0.$$

3.2 Vanishing of the z^1 Coefficient

We collect the z^1 coefficients from each term, factoring out $3x^2u^2$ to simplify.

1. From $u^3(Q_x R_y - Q_y R_x)$, the z^1 coefficient is:

$$u^3 [Q_{x,1} R_{y,0} - Q_{y,1} R_{x,0} - Q_{y,0} R_{x,1}].$$

Substituting the derivatives and factoring out $3x^2u^3$:

$$3x^2u^3 [u(3u-2)\psi'(u) - 2u(\psi(u) + (u-1)\psi'(u)) + 1 + 6(u-1)\varphi(u) + 3(u-1)^2\varphi'(u)].$$

We expand and simplify the bracket:

$$T_1 = u^2\psi'(u) - 2u\psi(u) + 1 + 6(u-1)\varphi(u) + 3(u-1)^2\varphi'(u).$$

So the first term contributes:

$$3x^2u^3 [u^2\psi'(u) - 2u\psi(u) + 1 + 6(u-1)\varphi(u) + 3(u-1)^2\varphi'(u)].$$

2. From $-3xu^2(P_x R_y - P_y R_x)$, the z^1 coefficient is:

$$-3xu^2 [P_{x,1} R_{y,0} - P_{y,1} R_{x,0} - P_{y,0} R_{x,1}].$$

Factoring out $9x^2u^2$:

$$-9x^2u^2 [-u^2\psi(u) + (u-1)(3u-1)\varphi(u) + u(u-1)^2\varphi'(u)].$$

3. From $-x^3(P_x Q_y - P_y Q_x)$, the z^1 coefficient is:

$$-x^3 [P_{x,1} Q_{y,0} + P_{x,0} Q_{y,1} - P_{y,1} Q_{x,0} - P_{y,0} Q_{x,1}].$$

Factoring out $3x^2u^2$:

$$3x^2u^2 [-(u-1) + 2(u-1)(2u-1)\varphi(u) + u(u-1)^2\varphi'(u)].$$

Summing these three parts and factoring out $3x^2u^2$, we obtain:

$$\begin{aligned} \frac{B}{3x^2u^2} &= u [u^2\psi'(u) - 2u\psi(u) + 1 + 6(u-1)\varphi(u) + 3(u-1)^2\varphi'(u)] \\ &\quad - 3 [-u^2\psi(u) + (u-1)(3u-1)\varphi(u) + u(u-1)^2\varphi'(u)] \\ &\quad + [-(u-1) + 2(u-1)(2u-1)\varphi(u) + u(u-1)^2\varphi'(u)]. \end{aligned}$$

Collecting terms in $\psi'(u)$, $\psi(u)$, $\varphi(u)$, $\varphi'(u)$ and the constant:

$$\frac{B}{3x^2u^2} = u^3\psi'(u) + u^2\psi(u) + (u^2-1)\varphi(u) + u(u-1)^2\varphi'(u) + 1.$$

Differentiating the Bézout identity (3) with respect to u :

$$2(u-1)\varphi(u) + (u-1)^2\varphi'(u) + 2u\psi(u) + u^2\psi'(u) = 1. \quad (9)$$

Multiplying this derivative relation by u :

$$u^3\psi'(u) = u - 2u^2\psi(u) - 2u(u-1)\varphi(u) - u(u-1)^2\varphi'(u).$$

Substituting this expression into the sum for $B/3x^2u^2$:

$$\begin{aligned}
\frac{B}{3x^2u^2} &= [u - 2u^2\psi(u) - 2u(u-1)\varphi(u) - u(u-1)^2\varphi'(u)] + u^2\psi(u) + (u^2-1)\varphi(u) + u(u-1)^2\varphi'(u) + 1 \\
&= u - u^2\psi(u) + [(u^2-1) - 2u(u-1)]\varphi(u) + 1 \\
&= u - u^2\psi(u) + [-u^2 + 2u - 1]\varphi(u) + 1 \\
&= u - u^2\psi(u) - (u-1)^2\varphi(u) + 1 \\
&= u - [u^2\psi(u) + (u-1)^2\varphi(u)] + 1.
\end{aligned}$$

Applying the Bézout identity (3):

$$\frac{B}{3x^2u^2} = u - (u+1) + 1 = 0.$$

Hence, $B = 0$.

3.3 Evaluation of the Constant Term C

Since $A = 0$ and $B = 0$, the determinant is constant in z . We evaluate it at $z = 0$ by direct substitution:

$$\det J_F = C = \det J_F \Big|_{z=0} = -2.$$

This completes the proof. $\square$

4 Analysis of the Fibers and Non-Injectivity

We show that the polynomial syzygies reduce the characterization of the fibers of $F_{t(u)}$ to a quadratic equation in u , replacing multivariate polynomial system solving. To disprove the Jacobian Conjecture, it is sufficient to show that the mapping is non-injective on a single fiber.

Theorem 4.1. *Let $\eta = (p, q, r)^T \in \mathbb{C}^3$ with $q = 0$. The polynomial syzygies (7) and (8) reduce the problem of determining the fiber $F_{t(u)}^{-1}(\eta)$ to solving a quadratic equation in the auxiliary variable $u = 1 + xy$.*

Proof. Let $(x, y, z) \in F_{t(u)}^{-1}(\eta)$, so that $P(x, y, z) = p$, $Q(x, y, z) = 0$, and $R(x, y, z) = r$. Substituting these values into (7) and (8) yields the system:

$$u(0 - y) = 3xp \implies uy = -3xp, \quad (10)$$

$$x^3p + u^3r = xu(u+1). \quad (11)$$

We analyze this system in two cases. If $x = 0$, then $u = 1$, and (10) yields $y = 0$. If $x \neq 0$, we can divide (11) by x :

$$x^2p + \frac{u^3r}{x} = u(u+1). \quad (12)$$

From (10), we multiply by x :

$$xyu = -3x^2p \implies (u-1)u = -3x^2p \implies x^2p = -\frac{1}{3}u(u-1).$$

Substituting this expression for x^2p into (12) when $r = 0$:

$$-\frac{1}{3}u(u-1) = u(u+1) \implies -\frac{1}{3}(u-1) = u+1.$$

This is a simple linear/quadratic equation in u , showing that the (x, y) -locus of the fiber remains completely rigid. $\square$

Corollary 4.2. *For any polynomial $t(u) \in \mathbb{C}[u]$, the fiber $F_{t(u)}^{-1}(\eta)$ of the point $\eta = (-\frac{1}{4}, 0, 0)$ consists of exactly three distinct points in $\mathbb{C}^3$:*

$$A = \left(0, 0, -\frac{1}{4}\right), \quad B = \left(1, -\frac{3}{2}, 2 - 9\varphi(-1/2)\right), \quad C = \left(-1, \frac{3}{2}, 2 - 9\varphi(-1/2)\right).$$

Proof. Let $(x, y, z) \in \mathbb{C}^3$ be a point in the fiber $F_{t(u)}^{-1}(\eta)$. Then we must have:

$$P(x, y, z) = -\frac{1}{4}, \quad Q(x, y, z) = 0, \quad R(x, y, z) = 0.$$

Substituting $p = -1/4$, $q = 0$, and $r = 0$ into the syzygy equations (10) and (11) yields:

$$u(0 - y) = 3x \left(-\frac{1}{4}\right) \implies uy = \frac{3}{4}x, \quad (13)$$

$$x^3 \left(-\frac{1}{4}\right) + u^3(0) = xu(u + 1) \implies -\frac{1}{4}x^3 = xu(u + 1). \quad (14)$$

4.1 Case 1: $x = 0$

If $x = 0$, then from (13) we obtain:

$$uy = 0 \implies (1 + xy)y = 0 \implies (1 + 0)y = 0 \implies y = 0.$$

With $x = 0$ and $y = 0$, the parameter is $u = 1$. Substituting these values into the equation $P(x, y, z) = -\frac{1}{4}$ yields:

$$u^3z + y^2u\varphi(u) = -\frac{1}{4} \implies z = -\frac{1}{4}.$$

This gives the unique solution:

$$A = \left(0, 0, -\frac{1}{4}\right).$$

4.2 Case 2: $x \neq 0$

If $x \neq 0$, we can divide both sides of (14) by x :

$$-\frac{1}{4}x^2 = u(u + 1). \quad (15)$$

From (13), we have $yu = \frac{3}{4}x$. Multiplying by x gives:

$$xyu = \frac{3}{4}x^2 \implies (u - 1)u = \frac{3}{4}x^2 \implies x^2 = \frac{4}{3}u(u - 1).$$

Substituting this expression for x^2 into (15):

$$-\frac{1}{4} \left(\frac{4}{3}u(u - 1) \right) = u(u + 1) \implies -\frac{1}{3}u(u - 1) = u(u + 1).$$

Since $x \neq 0$ and $x^2 = \frac{4}{3}u(u - 1)$, we must have $u \neq 0$ and $u \neq 1$. Dividing by u :

$$-\frac{1}{3}(u - 1) = u + 1 \implies -u + 1 = 3u + 3 \implies 4u = -2 \implies u = -\frac{1}{2}.$$

With $u = -\frac{1}{2}$, we compute x^2 :

$$x^2 = \frac{4}{3} \left(-\frac{1}{2} \right) \left(-\frac{3}{2} \right) = 1 \implies x = \pm 1.$$

We solve for y using $xy = u - 1 = -\frac{3}{2}$:

- If $x = 1$, then $y = -\frac{3}{2}$.
- If $x = -1$, then $y = \frac{3}{2}$.

Finally, we solve for z using $P(x, y, z) = -\frac{1}{4}$. We have $u = -\frac{1}{2}$ and $y^2 = \frac{9}{4}$:

$$u^3 z + y^2 u \varphi(u) = -\frac{1}{4} \implies \left(-\frac{1}{8}\right) z + \left(\frac{9}{4}\right) \left(-\frac{1}{2}\right) \varphi(-1/2) = -\frac{1}{4}.$$

Simplifying:

$$-\frac{1}{8} z - \frac{9}{8} \varphi(-1/2) = -\frac{1}{4} \implies z = 2 - 9\varphi(-1/2).$$

This yields exactly two solutions for $x \neq 0$:

$$B = \left(1, -\frac{3}{2}, 2 - 9\varphi(-1/2)\right) \quad \text{and} \quad C = \left(-1, \frac{3}{2}, 2 - 9\varphi(-1/2)\right).$$

Combining both cases, the fiber contains exactly the three points A, B , and C for any choice of $t(u)$. $\square$

5 Inequivalence and Moduli of Counterexamples under $\text{Aut}(\mathbb{C}^3)$

A fundamental question regarding the family of maps $F_{t(u)}$ is whether they are genuinely distinct up to coordinates, or whether they correspond to a single orbit under coordinate transformations.

We define two polynomial mappings $F_1, F_2 : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ to be equivalent under the action of the affine automorphism group $\text{Aff}(\mathbb{C}^3) \subset \text{Aut}(\mathbb{C}^3)$ if there exist affine automorphisms $\sigma, \tau \in \text{Aff}(\mathbb{C}^3)$ such that:

$$F_2 = \sigma \circ F_1 \circ \tau.$$

Composition with affine transformations on the right (variable substitution) and on the left (linear output mixing) behaves in a controlled manner with respect to degree.

Theorem 5.1. *The family of counterexamples $\{F_{t(u)}\}_{t(u) \in \mathbb{C}[u]}$ spans infinitely many distinct affine equivalence classes (orbits) under the action of $\text{Aff}(\mathbb{C}^3) \times \text{Aff}(\mathbb{C}^3)$.*

Proof. Let $F = F_{t(u)} = (P, Q, R)$. We compute the degrees of the coordinate polynomials P, Q, R in the variables x, y, z . Let $d = \deg(t(u))$. Since $u = 1 + xy$ has degree 2:

$$\begin{aligned} \deg(P) &= \max(\deg(u^3 z), \deg(y^2 u \varphi(u))) = \max(7, 2 + 2 + \deg(\varphi(u))) \\ &= \max(7, 4 + 2(2 + d)) = 8 + 2d. \end{aligned}$$

Similarly, for Q :

$$\deg(Q) = \max(\deg(3xu^2 z), \deg(3xy^2 \varphi(u))) = \max(6, 1 + 2 + (4 + 2d)) = 7 + 2d.$$

For R :

$$\deg(R) = \max(\deg(x\psi(u)), \deg(x^3 z)) = \max(1 + 2(2 + d), 4) = 5 + 2d.$$

These degrees are strictly ordered:

$$\deg(R) < \deg(Q) < \deg(P) = 8 + 2d.$$

Post-composition with an invertible linear automorphism σ corresponds to taking linear combinations of the output components P, Q, R . Since the degrees of P, Q , and R are strictly ordered, any non-trivial linear combination containing P must have degree exactly $8 + 2d$, as the leading terms of P cannot be canceled by those of the lower-degree polynomials Q and R . Since σ is invertible, it must contain a row with a non-zero coefficient for P . Thus, the maximum degree of the components of $\sigma \circ F_{t(u)} \circ \tau$ is preserved and remains exactly $8 + 2d$. As the degree of $t(u)$ can be chosen to be arbitrarily large, the family spans infinitely many distinct affine equivalence classes. $\square$

6 Discussion

The discovery of the Alpöge-Mathew counterexample settled the Jacobian Conjecture in the negative for all dimensions $n \geq 3$, leaving the $n = 2$ case as the final remaining open question.

6.1 Why the Counterexample is Surprising

For decades, search efforts focused on homogeneous polynomial maps of degree 3, following the Bass-Connell-Wright reduction. The Alpöge-Mathew map is not of this type: it has coordinate degrees $\deg(P) = 4$, $\deg(Q) = 5$, and $\deg(R) = 3$. This suggests that the degree reduction, while mathematically valid, may restrict the search space in a way that obscures simpler non-injective structures.

6.2 Significance of the Polynomial Syzygies and Rigidity

The presence of the syzygies (7) and (8) indicates that the coordinate functions are not algebraically independent over the ring $\mathbb{C}[x, y]$. We have shown that the family of counterexamples is infinite-dimensional and parameterized by the Bézout equation. Furthermore, the (x, y) -locus of the fiber over $(-1/4, 0, 0)$ is completely rigid under this parameterized family, indicating that the three-point collision is a structural invariant of the ansatz itself rather than a coincidence of specific constants.

6.3 Obstructions to a Two-Dimensional Analogue

A natural question is whether the syzygy-driven shear mechanism used in the three-dimensional counterexample family can be directly transplanted to construct two-dimensional counterexamples. In this subsection, we show that this exact mechanism encounters a structural algebraic obstruction in dimension $n = 2$.

Consider the most direct two-variable analogue of our ansatz, where the components P and Q are linear in y (mirroring how the components in the three-dimensional map are linear in z):

$$P(x, y) = u^2 y + f(x), \tag{16}$$

$$Q(x, y) = -x^2 y + g(x), \tag{17}$$

where $u = 1 + xy$, and $f(x), g(x) \in \mathbb{C}[x]$ are arbitrary univariate polynomials in x .

We compute the Jacobian determinant of this two-dimensional mapping:

$$\det J_F = P_x Q_y - P_y Q_x.$$

Substituting the partial derivatives of P and Q :

$$P_x = 2uy \frac{\partial u}{\partial x} + f'(x) = 2y^2(1 + xy) + f'(x) = 2xy^3 + 2y^2 + f'(x),$$

$$P_y = u^2 + 2uy \frac{\partial u}{\partial y} = (1 + xy)^2 + 2xy(1 + xy) = (1 + xy)(1 + 3xy),$$

$$Q_x = -2xy - x^2 y \frac{\partial u}{\partial x} + g'(x) = -x^2 y^2 - 2xy + g'(x),$$

$$Q_y = -x^2 - x^2 y \frac{\partial u}{\partial y} = -x^3 y - x^2.$$

Expanding the determinant as a polynomial in y :

$$\det J_F = 4x^3 y^3 + (6x^2 - 3x^2 g'(x))y^2 + (2x - 4xg'(x))y - (x^2 f'(x) + g'(x)).$$

Notice that the leading term of the Jacobian determinant as a polynomial in y is $4x^3 y^3$. Since $f(x)$ and $g(x)$ depend solely on x , their derivatives cannot contribute any terms that could cancel this leading term. Thus, the Jacobian determinant can never be a constant unless $x = 0$, which is impossible for a polynomial map.

The underlying reason for this failure is structural. In three dimensions, the Jacobian determinant is computed via a cofactor expansion into three separate 2×2 minors, weighted by the z -derivatives

$(u^3, 3xu^2, -x^3)$. This allows a three-way cancellation between the different terms, which we successfully controlled using the freedom of the Bézout identity (3). In contrast, a 2×2 Jacobian determinant has no cofactor expansion structure to provide internal redundancy; it is simply a single product-minus-product relation, leaving no "third leg" to absorb the extra degrees of freedom. This shows that the syzygy-driven shear mechanism cannot be directly transplanted to dimension $n = 2$, highlighting the fundamental algebraic differences between the two-dimensional and higher-dimensional cases.

6.4 A Critical-Point Obstruction Theorem in Two Dimensions

To further understand why the syzygy-driven construction fails in two variables, we establish a general critical-point obstruction theorem for mappings where one coordinate function factors purely through an auxiliary variable.

Theorem 6.1. *Let $u(x, y) \in \mathbb{C}[x, y]$ be a non-constant polynomial. Let $F = (P, Q) : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be a polynomial mapping such that one component factors purely through u , i.e., $Q(x, y) = G(u(x, y))$ for some univariate polynomial $G \in \mathbb{C}[u]$. If $u(x, y)$ has a critical point in $\mathbb{C}^2$ (a point (x_0, y_0) where $u_x(x_0, y_0) = u_y(x_0, y_0) = 0$), then the Jacobian determinant $\det J_F$ must vanish at (x_0, y_0) . Consequently, F cannot be a Keller mapping.*

Proof. Let $F = (P, Q)$ with $Q(x, y) = G(u(x, y))$. We compute the partial derivatives of Q :

$$Q_x = G'(u)u_x, \quad Q_y = G'(u)u_y.$$

The Jacobian determinant of F is given by:

$$\det J_F = P_x Q_y - P_y Q_x = P_x (G'(u)u_y) - P_y (G'(u)u_x) = G'(u) (P_x u_y - P_y u_x).$$

Now, let $(x_0, y_0) \in \mathbb{C}^2$ be a critical point of $u(x, y)$, so that $u_x(x_0, y_0) = 0$ and $u_y(x_0, y_0) = 0$. Evaluating the expression at (x_0, y_0) yields:

$$\det J_F(x_0, y_0) = G'(u(x_0, y_0)) (P_x(x_0, y_0) \cdot 0 - P_y(x_0, y_0) \cdot 0) = 0.$$

Since a Keller mapping must satisfy $\det J_F = c$ for some constant $c \in \mathbb{C} \setminus \{0\}$, the vanishing of the Jacobian determinant at any point in $\mathbb{C}^2$ implies that F cannot be a Keller mapping. $\square$

For the natural auxiliary variable used in our three-dimensional family, $u(x, y) = 1 + xy$, the partial derivatives are $u_x = y$ and $u_y = x$. The unique critical point of u in $\mathbb{C}^2$ is the origin $(0, 0)$. Applying Theorem 6.1, any polynomial mapping $(P, G(1 + xy))$ must have a vanishing Jacobian determinant at $(0, 0)$, and thus cannot disprove the two-dimensional Jacobian Conjecture.

This obstruction is strict when Q depends purely on u . If we allow Q to have explicit dependence on x beyond u (i.e., $Q(x, y) = G(x, u(x, y))$), the partial derivative Q_x picks up an extra term $\frac{\partial G}{\partial x}$ which does not vanish at critical points of u , breaking this direct vanishing argument. Investigating the limits of this generalization remains an open question for future work.

6.5 A Taxonomy of Approaches to the Jacobian Conjecture

In light of the Alpöge-Mathew counterexample in dimension three and the obstruction results in dimension two, we present a structured taxonomy of the primary mathematical strategies that have been deployed to either construct counterexamples or prove the conjecture.

6.5.1 Constructive Approaches

1. **Cubic-Homogeneous and Cubic-Linear Reductions:** Bass, Connell, and Wright [2], and independently Yagzhev [5], proved that any counterexample to the Jacobian Conjecture in dimension n can be reduced to a cubic-homogeneous mapping of the form $F(x) = x - H(x)$, where H is homogeneous of degree 3. However, this reduction may require passing to a larger number of variables $N > n$,

meaning the reduction is not dimension-preserving. Building on this, Drużkowski (1983) proved that any counterexample can be further reduced to a *cubic-linear* form:

$$F(x) = x - (Ax)^{\circ 3},$$

where A is a matrix and the power is taken component-wise. This reduces the conjecture to nilpotency conditions on matrices.

2. **Triangular Shears with Auxiliary Variables:** This strategy introduces an auxiliary polynomial $u(x, y)$ (or $u(x, y, z, \dots)$) and constructs coordinates linear in one of the variables, utilizing a Bézout-type identity to force a constant Jacobian determinant. This is the mechanism underlying Alpöge's counterexample and our generalized family. As established in Theorem 6.1, this mechanism faces a critical-point obstruction in dimension two.
3. **Rational and Anisotropic Auxiliary-Curve Constructions:** Sergey Pinchuk [4] constructed a smooth polynomial counterexample to the *Real* Jacobian Conjecture in $\mathbb{R}^2$ using rational parameterizations along a curve with points at infinity and anisotropic variable weighting. Whether an analogous complex construction is possible in $\mathbb{C}^2$ remains a major open question.
4. **Structured Computational Search:** Computational search over degree-bounded families of polynomial maps remains a primary discovery tool. By parameterizing large families of Keller maps and using computer algebra systems to test for injectivity, new counterexample candidates can be identified and subsequently verified.

6.5.2 Obstruction and Proof Strategies

1. **The Dixmier Conjecture and Weyl Algebra Quantization:** The Dixmier Conjecture asserts that any endomorphism of the Weyl algebra $A_n(\mathbb{C})$ is an automorphism. Tsuchimoto (2005) and Belov-Kanel and Kontsevich (2007) established a stable equivalence: the Jacobian Conjecture in dimension $2n$ implies the Dixmier Conjecture in rank n , which in turn implies the Jacobian Conjecture in dimension n . This stable equivalence connects the Jacobian Conjecture to algebraic quantum mechanics and noncommutative geometry.
2. **The Asymptotic and Non-Properness Locus:** Because a non-injective Keller map $F : \mathbb{C}^n \rightarrow \mathbb{C}^n$ cannot be proper, its asymptotic variety or non-properness locus S_F (the set of points where fiber coordinates escape to infinity) must be non-empty. Zbigniew Jelonek and others have studied the geometry of S_F , establishing topological constraints on the kinds of curves that can serve as the non-properness set in $\mathbb{C}^2$ (e.g., proving that S_F must contain non-rational curves).
3. **Valuation Theory and Boundary Dynamics:** Favre, Jonsson, and others have developed a framework to study the asymptotic behavior of polynomial maps at the boundary of $\mathbb{C}^2$ using valuations at infinity and tree dynamics. This approach yields algebraic invariants (such as key polynomials and asymptotic degrees) that can detect the presence of non-injective behaviors.
4. **Degree-Bound Sieves:** This strategy seeks to verify the Jacobian Conjecture for all polynomial mappings up to a bounded degree. For example, Wang's theorem verifies the conjecture for all maps in $\mathbb{C}^2$ of degree ≤ 2 . This establishes a minimum degree threshold for any potential counterexample in dimension two.

6.6 The Image and Vanishing Conjectures

The disproof of the 3D Jacobian Conjecture immediately implies that related formulations, such as the Image Conjecture (which posits that the image of a differential operator under certain conditions is a Mathieu subspace) and the Vanishing Conjecture, are also false in dimensions $n \geq 3$.

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